



An-Najah National University
Faculty of Engineering Information Technology
Computer Engineering Department

**Distributed and Operating
Systems Assignment**

Supervisor
Dr. Samer Arandi

Students Names:
1- Hadi Irshaid
2- Zahi Abu Shalbak

Cache Consistency

```
class LRUCache:
    def __init__(self, capacity=5):
        self.cache = OrderedDict() # OrderedDict to maintain insertion order
        self.capacity = capacity # Max number of items the cache can hold

    def get(self, key):
        """Get item from cache."""
        if key in self.cache:
            self.cache.move_to_end(key) # Move the accessed item to the end (most recently used)
            return self.cache[key]
        return None # Return None if the key is not in cache

    def put(self, key, value):
        """Put item in cache."""
        if key in self.cache:
            self.cache.move_to_end(key) # Move to end if already in cache
        self.cache[key] = value
        if len(self.cache) > self.capacity: # If cache exceeds capacity, remove the oldest item
            self.cache.popitem(last=False)

    def invalidate(self, key):
        """Invalidate (remove) an item from cache."""
        if key in self.cache:
            del self.cache[key]

# Initialize cache with a capacity of 10 items
cache = LRUCache(capacity=10)
# Load balancing state for catalog and order servers
catalog_replicas = ["http://localhost:5050", "http://localhost:5052"]
order_replicas = ["http://localhost:5061", "http://localhost:5062"]
catalog_index = 0
order_index = 0
```

When I send get request in the first time before caching the data:

Catalog:

The screenshot shows a web browser interface for a REST client. The URL bar displays `http://localhost:5050/query?item_number=2`. The request method is set to `GET`. The response status is `200 OK` with a response time of `30 ms` and a body size of `216 B`. The response body is displayed in JSON format:

```
{
  "price": 25,
  "quantity": 7,
  "title": "RPCs for Noobs"
}
```

From the cache:

The screenshot shows a web browser's developer tools interface. At the top, a GET request is shown to the URL `http://localhost:5050/query?item_number=2`. Below the URL bar, the 'Params' tab is active, displaying a table of query parameters:

Key	Value	Description
item_number	2	

Below the params table, the 'Body' tab is active, showing the response in JSON format:

```
{
  "price": 25,
  "quantity": 7,
  "title": "RPCs for Noobs"
}
```

The status bar at the bottom indicates a 200 OK response with a response time of 12 ms and a size of 216 B.

Q1: Compute the average response time (query/buy) of your new systems. What is the response time with and without caching?

for info: 30ms

for search: 12ms

Gateway:

GET

http://localhost:5000/query?item_number=2

Send

Params • Authorization Headers (6) Body Scripts Settings Cookies

Query Params

☒	Key	Value	Description	...	Bulk Edit
☒	item_number	2			
	Key	Value	Description		

Body Cookies Headers (5) Test Results 200 OK • 2.06 s • 216 B • Save Response ...

Pretty Raw Preview Visualize JSON

```
1 {
2   "price": 25,
3   "quantity": 7,
4   "title": "RPCs for Noobs"
5 }
```

From the cache:

GET

http://localhost:5000/query?item_number=2

Send

Params • Authorization Headers (6) Body Scripts Settings Cookies

Query Params

☒	Key	Value	Description	...	Bulk Edit
☒	item_number	2			
	Key	Value	Description		

Body Cookies Headers (5) Test Results 200 OK • 7 ms • 216 B • Save Response ...

Pretty Raw Preview Visualize JSON

```
1 {
2   "price": 25,
3   "quantity": 7,
4   "title": "RPCs for Noobs"
5 }
```

The first time the cache memory was empty the second time it was located in the cache memory:

```
Terminal  Local x  Local (2) x  Local (3) x  +  v
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://192.168.1.19:5000
Press CTRL+C to quit
127.0.0.1 - - [18/Nov/2024 00:23:36] "GET /query?item_number=2 HTTP/1.1" 200 -
Cache hit for item_number 2
127.0.0.1 - - [18/Nov/2024 00:24:33] "GET /query?item_number=2 HTTP/1.1" 200 -
```

 http://localhost:5061/purchase/1 Save Share

POST

http://localhost:5061/purchase/1

Send

Params

Authorization

Headers (7)

Body

Scripts

Settings

Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body

Cookies

Headers (5)

Test Results

200 OK

9.09 s

209 B



 Save Response

...

Pretty

Raw

Preview

Visualize

JSON

...





```
1 {
2   "message": "Successfully purchased item 1"
3 }
```

HTTP <http://localhost:5062/purchase/8> Save Share

POST <http://localhost:5062/purchase/8> Send

Params Authorization Headers (7) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies Headers (5) Test Results **404 NOT FOUND** • 2.06 s • 217 B • Save Response ...

Pretty Raw Preview Visualize JSON

```
1 {  
2   "message": "Failed to fetch catalog item 8"  
3 }
```

With replica :

HTTP <http://localhost:5062/purchase/1> Save Share

POST <http://localhost:5062/purchase/1> Send

Params Authorization Headers (7) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies Headers (5) Test Results **200 OK** • 8.25 s • 209 B • Save Response ...

Pretty Raw Preview Visualize JSON

```
1 {  
2   "message": "Successfully purchased item 1"  
3 }
```

HTTP <http://localhost:5050/update> Save Share

PATCH <http://localhost:5050/update> Send

Params Authorization Headers (8) **Body** Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** Beautify

```
1 {"item_number":2,"stock_count":8}
```

Body Cookies Headers (5) Test Results **200 OK** • 4.10 s • 209 B • Save Response

Pretty Raw Preview Visualize **JSON** Copy Search

```
1 {
2   "message": "Updated record 2 successfully"
3 }
```

With replica :

HTTP <http://localhost:5051/update> Save Share

PATCH <http://localhost:5051/update> Send

Params Authorization Headers (8) **Body** Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** Beautify

```
1 {"item_number":2,"stock_count":8}
```

Body Cookies Headers (5) Test Results **200 OK** • 4.09 s • 209 B • Save Response

Pretty Raw Preview Visualize **JSON** Copy Search

```
1 {
2   "message": "Updated record 2 successfully"
3 }
```

Q2: How much does caching help?

Answers

for catalog server: 12ms, 30/12 (2.5 Faster than without cache)

for gateway server: 7ms, 2.06s/7ms (294 Faster than without using cache)

Invalidate Message

The image displays two screenshots of a REST client interface, likely Postman, showing the results of HTTP requests to a catalog server. Both requests are POST requests to the endpoint `http://localhost:5062/purchase/8` (top) and `http://localhost:5061/purchase/9` (bottom). The responses are 404 NOT FOUND, indicating that the catalog items do not exist. The response body for both requests is a JSON object: `{ "message": "Failed to fetch catalog item 8" }` (top) and `{ "message": "Failed to fetch catalog item 9" }` (bottom). The status bar shows a response time of 2.06 s and a body size of 217 B.

Top Screenshot: Request to `http://localhost:5062/purchase/8`. Response: 404 NOT FOUND. Body: `{ "message": "Failed to fetch catalog item 8" }`.

Bottom Screenshot: Request to `http://localhost:5061/purchase/9`. Response: 404 NOT FOUND. Body: `{ "message": "Failed to fetch catalog item 9" }`.

for Original services:

HTTP http://localhost:5000/query?item_number=1 Save Share

GET http://localhost:5000/query?item_number=1 Send

Params Authorization Headers (6) Body Scripts Settings Cookies

Query Params

☒	Key	Value	Description	...	Bulk Edit
☒	item_number	1			
	Key	Value	Description		

Body Cookies Headers (5) Test Results 500 INTERNAL SERVER ERROR • 4.24 s • 458 B • Save Response ...

Pretty Raw Preview Visualize HTML

```
1 <!doctype html>
2 <html lang=en>
3 <title>500 Internal Server Error</title>
4 <h1>Internal Server Error</h1>
5 <p>The server encountered an internal error and was unable to complete your request. Either the server is overloaded or there
  is an error in the application.</p>
6
```

HTTP http://localhost:5000/query?item_number=1 Save Share

GET http://localhost:5000/query?item_number=1 Send

Params Authorization Headers (6) Body Scripts Settings Cookies

Query Params

☒	Key	Value	Description	...	Bulk Edit
☒	item_number	1			
	Key	Value	Description		

Body Cookies Headers (5) Test Results 200 OK • 2.07 s • 253 B • Save Response ...

Pretty Raw Preview Visualize JSON

```
1 {
2   "price": 96,
3   "quantity": 50,
4   "title": "How to get a good grade in DOS in 40 minutes a day"
5 }
```

 http://localhost:5000/query?item_number=1

SaveShare

GET

http://localhost:5000/query?item_number=1

Send

Params

Authorization

Headers (6)

Body

Scripts

Settings

Cookies

Query Params

☒	Key	Value	Description	...	Bulk Edit
☒	item_number	1			
	Key	Value	Description		

Body

Cookies

Headers (5)

Test Results

200 OK • 7 ms • 253 B • Save Response

Pretty

Raw

Preview

Visualize

JSON

```
1 {
2   "price": 96,
3   "quantity": 50,
4   "title": "How to get a good grade in DOS in 40 minutes a day"
5 }
```

For Replica Service

The screenshot shows a REST client interface with a history bar at the top containing several requests. The active request is a GET to `http://localhost:5051/query?item_number=1`. Below the URL bar, the 'Params' tab is selected, showing a table with query parameters. The 'Body' tab is also visible, displaying a JSON response in 'Pretty' format. The status bar indicates a '200 OK' response with a 10 ms duration and 253 B of data.

GET `http://localhost:5051/query?item_number=1` **Send**

Params • Authorization Headers (6) Body Scripts Settings Cookies

Query Params

☒	Key	Value	Description	...	Bulk Edit
☒	item_number	1			
	Key	Value	Description		

Body Cookies Headers (5) Test Results **200 OK** • 10 ms • 253 B • Save Response ...

Pretty Raw Preview Visualize JSON

```
1 {
2   "price": 96,
3   "quantity": 50,
4   "title": "How to get a good grade in DOS in 40 minutes a day"
5 }
```

For gateway Replica

The screenshot shows a REST client interface with a history bar at the top. The active request is a GET to `http://localhost:5009/query?item_number=1`. Below the URL bar, the 'Params' tab is selected, showing a table with query parameters. The 'Body' tab is also visible, displaying a JSON response in 'Pretty' format. The status bar indicates a '200 OK' response with a 2.07 s duration and 253 B of data.

GET `http://localhost:5009/query?item_number=1` **Send**

Params • Authorization Headers (6) Body Scripts Settings Cookies

Query Params

☒	Key	Value	Description	...	Bulk Edit
☒	item_number	1			
	Key	Value	Description		

Body Cookies Headers (5) Test Results **200 OK** • 2.07 s • 253 B • Save Response ...

Pretty Raw Preview Visualize JSON

```
1 {
2   "price": 96,
3   "quantity": 50,
4   "title": "How to get a good grade in DOS in 40 minutes a day"
5 }
```

Part2: Dockerize your Application (Optional part):

I constructed my project Dockerized from scratch, i create docker container (image) for each service, and each service has its own port, and I use a volume for sharing data between the Guest OS (Docker Containers) & Host OS, and I created a Docker-compose file to run all containers at the same time with 1 simple command.

My .Yaml **file**:

```
version: '3.8'

services:
  catalog_api:
    build:
      context: ./catalog
    ports:
      - "5050:5000"
    networks:
      - app_network

  purchase_api:
    build:
      context: ./purchase
    ports:
      - "5060:5000"
    networks:
      - app_network

  gateway_api:
    build:
      context: ./gateway
    ports:
      - "8080:5000"
    networks:
      - app_network
    depends_on:
      - catalog_api
      - purchase_api

  load_balancer_api:
    build:
      context: ./loadbalancer
    ports:
      - "8090:4000"
    networks:
      - app_network
    depends_on:
      - catalog_api
      - purchase_api
```

```
networks:
  app_network:
    driver: bridge
```

Docker Containers(images) while running:

```
time="2024-11-18T00:46:20+02:00" level=warning msg="C:\\Users\\User\\Desktop\\library_microservices-main\\library_microservices-main\\compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 4/4[+] Running 2/4
 ✓ Container library_microservices-main-purchase_api-1      Started      1.7s
 ✓ Container library_microservices-main-catalog_api-1       Started      1.7s
 ✓ Container library_microservices-main-load_balancer_api-1  Started      4.6s
 ✓ Container library_microservices-main-gateway_api-1       Started      4.3s
```